

Online Internship on Open-Source Processor Architecture (IBM OpenPOWER) and SoC Design

Assignment No. 01: Combinational Logic Design

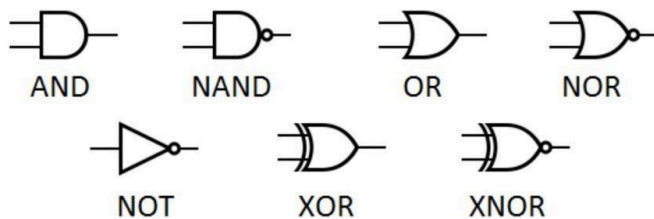
Instructions to Students

Please upload the complete compiled code, simulation screenshots, and documentation compiled as a single PDF file.

Assignment Questions

Q1. Design and Simulate all the six fundamental logic gates OR, NOT, NAND, NOR, XOR, XNOR in Vivado/Modelsim in dataflow and structural modeling styles.

Circuit / Block Diagram



Q2. Design a 2:1 Multiplexer using Verilog HDL in different modeling styles.

- a) Dataflow Modeling using continuous assignment (assign) statement.
- a) Dataflow Modeling using the conditional (?:) operator.
- a) Behavioral Modeling using always block.
- a) Behavioral Modeling using if-else statements.
- a) Behavioral Modeling using case statements.
- a) Gate-Level Modeling using logic primitives (and, or, not).